

Integration Theory MATM39: Summary of Important Results

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Chapter 1

Definition 1. Let X be an arbitrary set. A collection $\mathcal{A}$ of subsets of X is called a σ -algebra on X if:

- (a) $X \in \mathcal{A}$,
- (b) if $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$,
- (c) if $\{A_i\}_{i=1}^\infty \subset \mathcal{A}$, then $\bigcup_{i=1}^\infty A_i \in \mathcal{A}$,
- (d) if $\{A_i\}_{i=1}^\infty \subset \mathcal{A}$, then $\bigcap_{i=1}^\infty A_i \in \mathcal{A}$.

Proof. Assume (a), (b), and (c). Let $\{A_i\}_{i=1}^\infty \subset \mathcal{A}$. Then $A_i^c \in \mathcal{A}$ for all i , and by (c), $\bigcup_{i=1}^\infty A_i^c \in \mathcal{A}$. By (b) again, $(\bigcup_{i=1}^\infty A_i^c)^c \in \mathcal{A}$. By De Morgan's law,

$$\bigcap_{i=1}^\infty A_i = \left(\bigcup_{i=1}^\infty A_i^c \right)^c \in \mathcal{A}$$

□

Proposition 1. Let X be a set. Then the intersection of an arbitrary nonempty collection of σ -algebras on X is a σ -algebra on X .

Proof. Let $\mathcal{C}$ be a nonempty collection of σ -algebras on X , and define

$$\mathcal{A} := \bigcap_{\mathcal{B} \in \mathcal{C}} \mathcal{B}.$$

Since every $\mathcal{B} \in \mathcal{C}$ contains X , we have $X \in \mathcal{A}$. If $A \in \mathcal{A}$, then $A \in \mathcal{B}$ for every $\mathcal{B} \in \mathcal{C}$. Since each $\mathcal{B}$ is a σ -algebra, $A^c \in \mathcal{B}$ for every $\mathcal{B} \in \mathcal{C}$, hence $A^c \in \mathcal{A}$. If $\{A_i\}_{i=1}^\infty \subset \mathcal{A}$, then $\{A_i\}_{i=1}^\infty \subset \mathcal{B}$ for every $\mathcal{B} \in \mathcal{C}$. Hence

$$\bigcup_{i=1}^\infty A_i \in \mathcal{B}$$

for every $\mathcal{B} \in \mathcal{C}$, and therefore

$$\bigcup_{i=1}^\infty A_i \in \mathcal{A}.$$

Thus $\mathcal{A}$ is a σ -algebra on X .

□

Corollary 1. *Let X be a set, and let $\mathcal{F}$ be a family of subsets of X . Then there exists a smallest σ -algebra on X that contains $\mathcal{F}$.*

Proof. Let $\mathcal{C}$ be the collection of all σ -algebras on X that contain $\mathcal{F}$. This collection is nonempty, since $\mathcal{P}(X)$ is a σ -algebra on X and contains $\mathcal{F}$.

Define

$$\sigma(\mathcal{F}) := \bigcap_{\mathcal{A} \in \mathcal{C}} \mathcal{A}.$$

By the previous proposition, $\sigma(\mathcal{F})$ is a σ -algebra on X . It contains $\mathcal{F}$, since every $\mathcal{A} \in \mathcal{C}$ contains $\mathcal{F}$. Moreover, if $\mathcal{B}$ is any σ -algebra on X that contains $\mathcal{F}$, then $\mathcal{B} \in \mathcal{C}$, and therefore

$$\sigma(\mathcal{F}) \subset \mathcal{B}.$$

Thus $\sigma(\mathcal{F})$ is the smallest σ -algebra on X containing $\mathcal{F}$. $\square$

Definition 2. *The Borel σ -algebra on $\mathbb{R}^d$ is the σ -algebra generated by the open subsets of $\mathbb{R}^d$. It is denoted by $\mathcal{B}(\mathbb{R}^d)$. Its elements are called Borel sets.*

Proposition 2. *The Borel σ -algebra $\mathcal{B}(\mathbb{R})$ is generated by each of the following collections:*

- (a) *the collection of all closed subsets of $\mathbb{R}$,*
- (b) *the collection of all intervals of the form $(-\infty, b]$,*
- (c) *the collection of all intervals of the form $(a, b]$.*

Proof. Let $\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3$ denote the σ -algebras generated by the collections in (a), (b), and (c), respectively.

Since $\mathcal{B}(\mathbb{R})$ contains all open sets and is closed under complements, it contains all closed sets. Hence

$$\mathcal{B}(\mathbb{R}) \supset \mathcal{B}_1.$$

Every set $(-\infty, b]$ is closed, so $\mathcal{B}_1 \supset \mathcal{B}_2$. Moreover,

$$(a, b] = (-\infty, b] \cap (-\infty, a]^c,$$

so $\mathcal{B}_2 \supset \mathcal{B}_3$.

Finally, let $G \subset \mathbb{R}$ be open. For each $x \in G$, since G is open, there exist rational numbers $a, b \in \mathbb{Q}$ such that

$$x \in (a, b] \subset G.$$

Hence

$$G = \bigcup_{\substack{a, b \in \mathbb{Q} \\ (a, b] \subset G}} (a, b].$$

This is a countable union, since $\mathbb{Q}^2$ is countable. Therefore $G \in \mathcal{B}_3$. Thus every open set belongs to $\mathcal{B}_3$, so

$$\mathcal{B}(\mathbb{R}) \subset \mathcal{B}_3.$$

Therefore all four σ -algebras are equal. $\square$

Chapter 2

Chapter 3

Chapter 5